

SKYGUARD AI

Self-Healing Weather Observation Network

Intelligent Real-Time Anomaly Detection for T, P, RH Sensors in Automatic Weather Stations

Problem Statement: AI/ML-Based Intelligent Anomaly Detection for Automatic Weather Stations (AWS)

Theme: Smart Automation & Observational Networks | **Category:** Software (with Edge-AI hardware)

Parameters: Temperature °C • Atmospheric Pressure hPa • Relative Humidity %

Team ID: _____ | **Team Name:** _____ | **Mentor:** _____

The Problem

- AWS data feeds forecasts, aviation, farming, disaster warnings — but is polluted by sensor faults, spikes, frozen values, dropout, drift, comms errors.
- Traditional threshold-only QC misses complex / hidden / multivariate anomalies.
- A single detector **cannot tell a broken sensor from a real heatwave.**

Our Solution — in one line

A 6-layer edge+cloud AI fusion pipeline mirroring WMO L0-L5 QC, with SHAP explainability and a self-aware, self-healing sensor-health loop.

The Core Insight

If only Delhi reports 55°C → **broken sensor**. If Delhi *and* Lucknow *and* Jaipur all report 55°C → **real heatwave, alert don't fix**. Multi-layer + neighbour consensus separates fault from weather.

The Fusion Matrix (the brain)

	Neighbours AGREE ($Z < 3.1$)	Neighbours DISAGREE ($Z \geq 3.1$)
Model flagged (L1-L4)	REGIONAL_EVENT real weather — alert, not fault — Medium	SENSOR_FAULT sensor is broken — High
No model flag	HEALTHY all clear — no action	CHECK_HEALTH subtle drift — schedule maintenance — Low

Novelty Bullets

- **Self-healing loop:** composite health score + predictive time-to-failure + auto-imputation — direct answer to the grand challenge.
- **Edge-deployed ML on ESP32-S3** — no competitor does per-station ML inference.
- **Full physics + ML + foundation-forecast + spatial fusion** — not published as an integrated system.
- **Transmit-only-on-suspicion** → 72% energy saving, ~10 mo battery vs ~3 mo.
- **Explainable by construction** — SHAP + per-fault reason + confidence on every alert.

Pipeline (Edge → Cloud → Fusion)



The 6 Layers

#	Layer	Role	WMO
L1	Physics Sanity	Range, dew-point, barometric, rate	L1-L2
L2	ML	Ext. IsoForest + SHAP + LSTM-AE + trend	L2
L3	Frozen	std < 0.01 / d²x/dt² ≈ 0	L2
L4	Forecasting	Chronos-2 corridor breach	L5
L5	Spatial	Robust Z (median/MAD) vs Haversine neighbours	L3
L6	Fusion	2×2 matrix → verdict + confidence + root cause	L4+

Self-Healing Sensor Health Score [0,1]

Health = $w_{sub1} \cdot (1 - \text{bias}) + w_{sub2} \cdot (1 - \text{residual_rms}) + w_{sub3} \cdot \text{availability} + w_{sub4} \cdot (1 - \text{anomaly_rate}) + w_{sub5} \cdot (1 - \text{age_degradation})$

Tiers: ≥0.85 Green / 0.6–0.85 Amber / 0.4–0.6 Orange / <0.4 Red. Predict weeks-to-failure by trending the score → flag before bad data flows downstream → auto-upweight imputed values until fixed = **trustworthy data under all conditions**.

Tech Stack

Tier	Technology
Edge	ESP32-S3, emlearn (IsoForest <10KB), TFLite-Micro (int8 autoencoder)
Gateway	Node.js + Fastify, WebSocket
ML core	Python + FastAPI; scikit-learn (EIF), SHAP, LSTM-AE, amazon/chronos-2, alibi-detect
Store	TimescaleDB (hypertable) + Redis (windows + neighbour map)
Queue	BullMQ on Redis → horizontal scale
Frontend	React + Leaflet (health-colour map) + Recharts (SHAP, corridor)

Data Sources

- **Open-Meteo Archive API** (ERA5, free, no key, hourly 1940–now)
- **NOAA ISD** (real station-level noise, public domain)
- **De Bruijn 2016** injected-fault benchmark (5.78M labelled pts)
- **IMD AWS API** (api.imd.gov.in) — production integration path
- **WMO OSCAR/Surface** — station metadata for neighbour map

Key Thresholds (sourced)

Check	Threshold	Source
Temp rate (fault)	$ \Delta T > 19.4 \text{ }^\circ\text{C/hr}$	NOAA MADIS (35°F/hr)
Temp rate (alert)	$ \Delta T > 8 \text{ }^\circ\text{C/hr}$	Tightened watch level
Pressure rate (fault)	$ \Delta P > 15 \text{ hPa/hr}$	NOAA MADIS sea-level
Pressure rate (alert)	$ \Delta P > 3 \text{ hPa/hr}$	Cyclone-class (weather, not fault)
	$ \Delta RH > 50 \text{ \% /hr}$	NOAA MADIS

Feasibility

- **Standards-aligned:** 6 layers mirror WMO L0–L5 (CIMO Guide WMO-No. 8; MADIS; WDQMS) — not invented in a vacuum.
- **Free, no-auth data:** Open-Meteo/ERA5 + NOAA ISD — no access blocker.
- **Cheap hardware:** ESP32-S3 ≈ ₹500; commodity cloud.
- **Each component literature-validated:** ECMWF LSTM-AE (operational 2023); Met Éann LSTM-AE 99.6% (2026); IEEE edge-IsoForest <10ms (2026); Kataria 6-layer F1=0.91 (2026).
- **Maturity ladder:** Idea → Pilot (1 region, 50 stations, IMD API) → Production (675+ IMD, K8s, audit logs, WMO traceability).

Risks & Mitigations

Risk	Mitigation
Isolated stations (no buddies)	Multi-layer agreement + NWP O-B “virtual buddy” (ECMWF)
IsoForest blind spots (clusters, drift, ghost clusters)	Extended IsoForest + LSTM-AE sibling + PSI-driven retrain
ML flags real weather as anomaly	L5 spatial consensus → REGIONAL_EVENT; de-seasonalize inputs (ECMWF)
Class imbalance <5%	Eval with PR-AUC / F2, never raw accuracy
ESP32 RAM / OTA / ADC noise	int8 quant, OTA rollback, calibration + filter
Regulatory / black-box	SHAP + human-in-loop + deterministic fallback + audit log (WMO Congress Oct 2025)
No telemetry integrity standard	HMAC-signed readings + tamper classifier (closes a WMO gap)

Sustainability

72% energy saving

~10 mo battery vs ~3 mo

Less e-waste

Fewer field trips

Target Audience

- IMD & State met departments (675+ AWS)
- Aviation, agriculture, disaster management
- Climate research, NWP data assimilation
- Every national met service globally (NOAA ISD 20,000+ stations)

Scale of Impact

675+ IMD stations

20,000+ NOAA scale

3 parameters → global applicability

Benefits

Type	Benefit
Social	Better forecasts → better disaster warnings → lives saved
Economic	Less bad-data propagation → fewer wrong forecasts → lower economic loss; longer sensor life → lower maintenance cost
Environmental	72% energy saving + longer field life → less e-waste, fewer truck rolls
Operational	Predictive maintenance → fix sensors <i>before</i> bad data, not after

User Experience — Every Alert Explained

Alert card = verdict + confidence badge + **blamed sensor (SHAP)** + plain-English reason + recommended action + imputed value + source.

Example: “SENSOR_FAULT (HIGH) — pressure_hpa blamed by SHAP (1.8). Sea-level pressure at 2205m altitude. Action: recalibrate barometer. Imputed P ≈ 780 hPa (neighbour median).”

Dashboard

- **Leaflet map** — stations coloured by health (green/amber/orange/red)
- **Live alert feed** — colour-coded severity, expandable per-layer
- **SHAP bar chart** — per-feature contribution for the flagged reading
- **Forecast corridor chart** — p5-p95 band vs actual (red if breach)
- **Sensor-health panel** — rolling score + trend + predicted time-to-failure

Expected Outputs (per problem statement)

Real-time alerts • severity + confidence • root-cause class • visualization dashboard • sensor health status • corrected-data estimation (optional) — **all delivered.**

Academic & Production-QC References

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